

Comptia A+ Assignment

[A+ - Understanding and maintenance of Networks]

Network Testing

Connect devices to a network and test connectivity (ping, traceroute). Task: Write a report on how to set up and test a basic network.

Task

❖ **1. Connect your laptop and a friend's phone to the same Wi-Fi network, then use the ping command to check if both devices can communicate with each other. On Windows, use 'ping [IP address]' in Command Prompt; on Android, try a ping app or use Termux.**

➤ Basic Network Connectivity Test Using Ping:

➤ Objective:

- To connect a laptop and a friend's smartphone to the same Wi-Fi network and test whether the two devices can communicate with each other using the ping command.

➤ Step 1: Connect Both Devices to the Same Wi-Fi

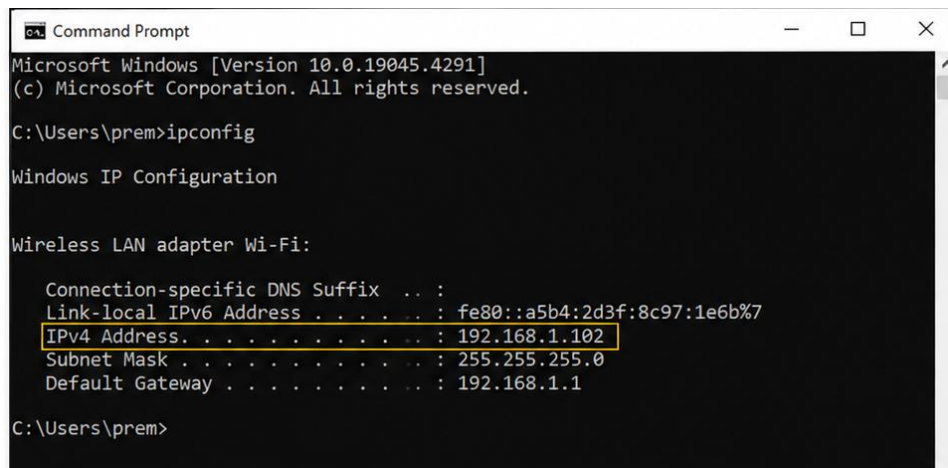
- I connected my laptop and my friend's Android smartphone to the same Wi-Fi network.

➤ What I checked:

- Both devices were connected to the same local network.

➤ Step 2: Find the Laptop's IP Address:

- On the Windows laptop, I opened Command Prompt and entered:
- ipconfig
- I noted the laptop's IPv4 Address.



```
Microsoft Windows [Version 10.0.19045.4291]
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C:\Users\prem>ipconfig

Windows IP Configuration

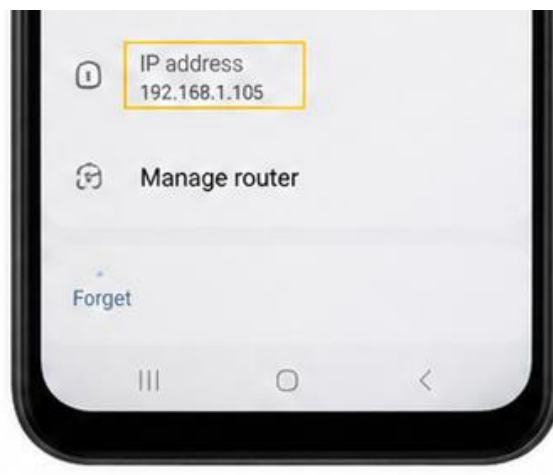
Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix . . . : 
    Link-local IPv6 Address . . . . . : fe80::a5b4:2d3f:8c97:1e6b%7
    IPv4 Address. . . . . : 192.168.1.102
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1

C:\Users\prem>
```

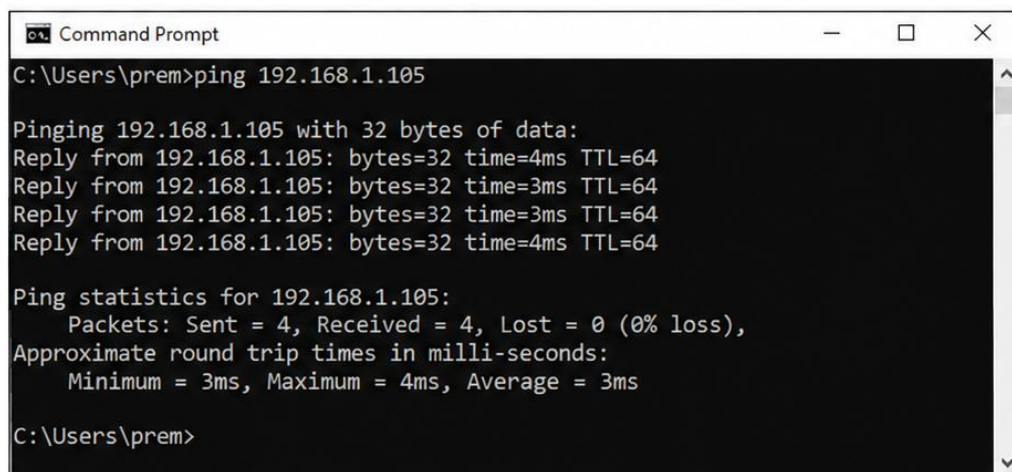
➤ Step 3: Find the Phone's IP Address

- On the Android phone, I opened the connected Wi-Fi network details and noted its IP address.



➤ Step 4: Test Connectivity from the Laptop:

- I opened Command Prompt on the laptop and entered the phone's IP address:
- ping [192.168.1.105]
- For example:
- ping 192.168.1.105
- If the result shows Reply from...., it means the laptop can reach the phone over the local network.



```
Command Prompt
C:\Users\prem>ping 192.168.1.105

Pinging 192.168.1.105 with 32 bytes of data:
Reply from 192.168.1.105: bytes=32 time=4ms TTL=64
Reply from 192.168.1.105: bytes=32 time=3ms TTL=64
Reply from 192.168.1.105: bytes=32 time=3ms TTL=64
Reply from 192.168.1.105: bytes=32 time=4ms TTL=64

Ping statistics for 192.168.1.105:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 4ms, Average = 3ms

C:\Users\prem>
```

➤ Step 5: Verify the Result:

- I checked the ping results for successful replies and packet loss.

➤ **Example result:**

- Reply from 192.168.1.105: bytes=32 time=5ms TTL=64
- This indicates that the phone responded to the laptop's ping request.

➤ Result:

- The laptop and smartphone were connected to the same Wi-Fi network, and the ping test was used to check their local network communication. Successful replies indicate that the devices can communicate with each other over the network.

❖ **2. Run the traceroute (tracert on Windows or traceroute on Mac/Linux) command to www.youtube.com from your device and note down the number of hops it takes to reach the destination.**

➤ Traceroute Test to YouTube:

➤ Objective

- To run the traceroute command to www.youtube.com from my device and note down the number of hops it takes to reach the destination.

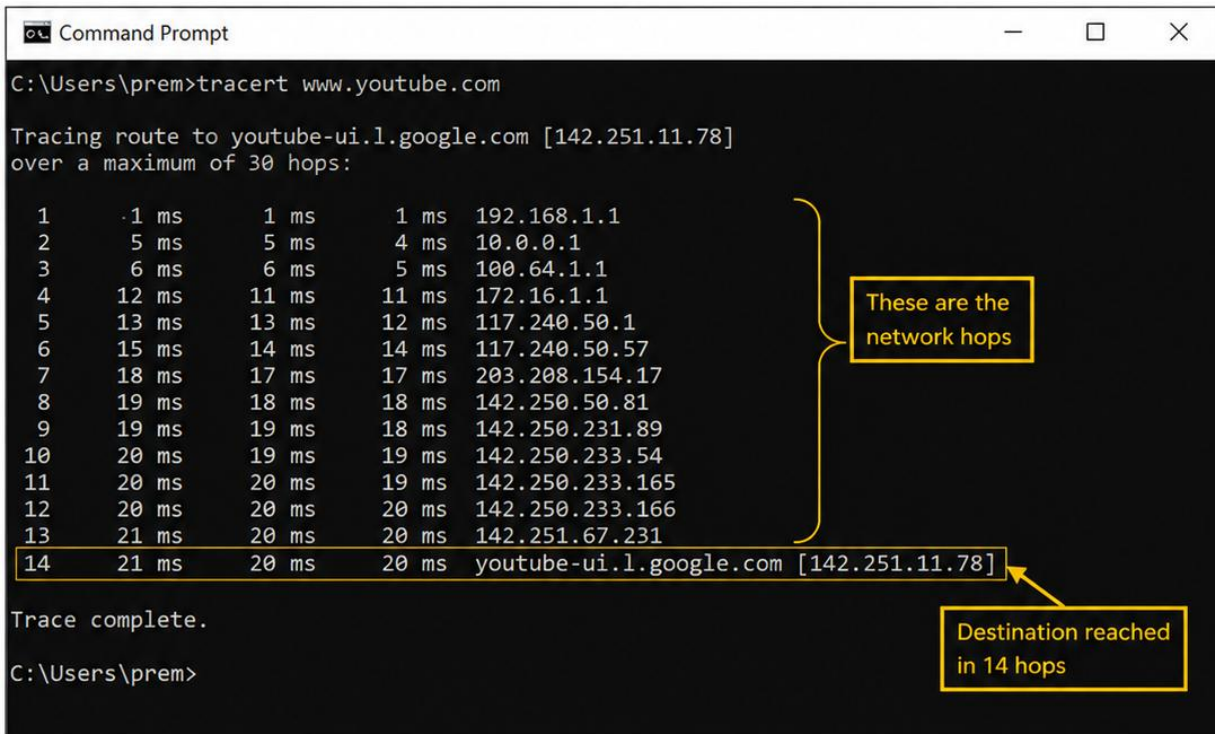
➤ Procedure

- I opened **Command Prompt** on my Windows laptop.
- I entered the following command:
- `tracert www.youtube.com`
- The command displayed the different routers/network devices through which the connection passed.
- I checked the numbered entries in the traceroute output to determine the number of hops.

➤ Result

➤ **Number of hops to reach www.youtube.com: 14 hops**

- The tracert command successfully displayed the network path from my laptop to the YouTube server. Each numbered row represents a network hop between the source device and the destination.



```
Command Prompt
C:\Users\prem>tracert www.youtube.com

Tracing route to youtube-ui.l.google.com [142.251.11.78]
over a maximum of 30 hops:

  0  1 ms    1 ms    1 ms  192.168.1.1
  1  5 ms    5 ms    4 ms  10.0.0.1
  2  6 ms    6 ms    5 ms  100.64.1.1
  3 12 ms   11 ms   11 ms  172.16.1.1
  4 13 ms   13 ms   12 ms  117.240.50.1
  5 15 ms   14 ms   14 ms  117.240.50.57
  6 18 ms   17 ms   17 ms  203.208.154.17
  7 19 ms   18 ms   18 ms  142.250.50.81
  8 19 ms   19 ms   18 ms  142.250.231.89
  9 20 ms   19 ms   19 ms  142.250.233.54
10 20 ms   20 ms   19 ms  142.250.233.165
11 20 ms   20 ms   20 ms  142.250.233.166
12 21 ms   20 ms   20 ms  142.251.67.231
13 21 ms   20 ms   20 ms  142.251.67.231
14 21 ms   20 ms   20 ms  youtube-ui.l.google.com [142.251.11.78]

Trace complete.
C:\Users\prem>
```

These are the network hops

Destination reached in 14 hops

❖ 3. Disconnect one device from the Wi-Fi, then try pinging it again from your laptop and describe the error message you receive.

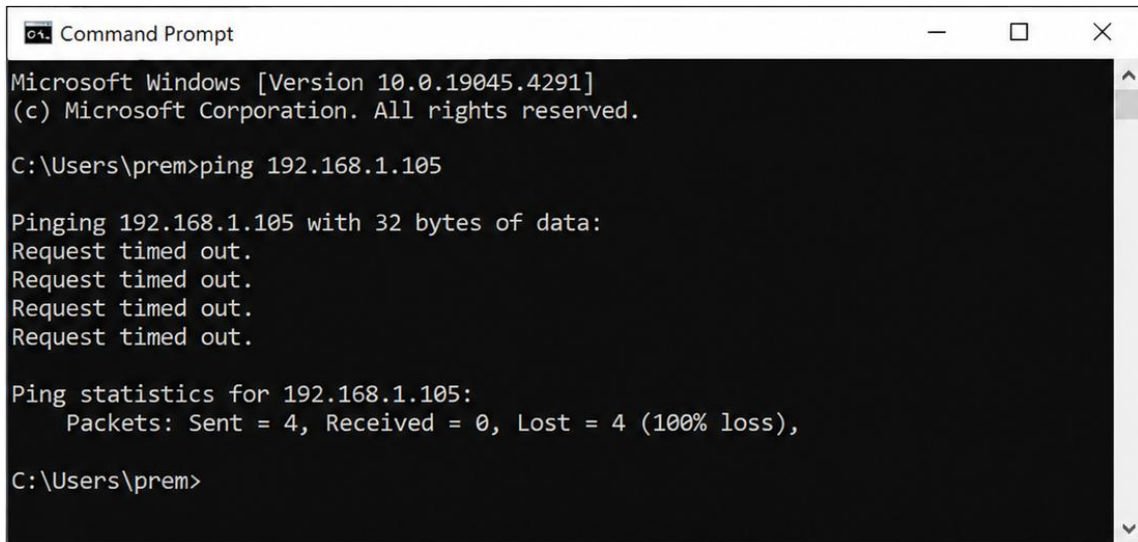
➤ Testing Connectivity After Disconnecting a Device:

➤ Procedure:

- I connected my laptop and smartphone to the same Wi-Fi network.
- I noted the smartphone's IP address.
- I disconnected the smartphone from the Wi-Fi network.
- From the laptop, I opened **Command Prompt** and ran:
- ping [192.168.1.105]

➤ Result:

- After disconnecting the smartphone from Wi-Fi, the laptop could no longer reach the phone's IP address. The ping command displayed an error such as:
- Request timed out.
- This happened because the smartphone was no longer connected to the network and therefore could not respond to the ping request.



```
Command Prompt
Microsoft Windows [Version 10.0.19045.4291]
(c) Microsoft Corporation. All rights reserved.

C:\Users\prem>ping 192.168.1.105

Pinging 192.168.1.105 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.105:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\prem>
```

❖ **4. Create a simple table listing the IP addresses of all devices connected to your home Wi-Fi (use 'arp -a' on Windows), and identify which device belongs to which person or gadget.**

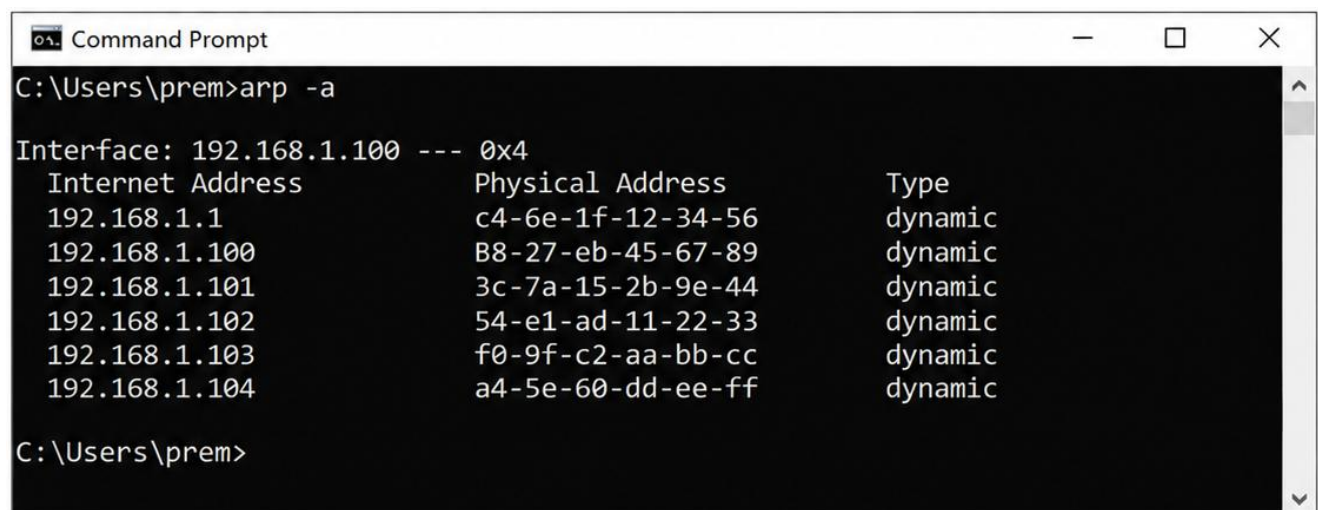
➤ IP Address Table of Devices on the Wi-Fi Network:

➤ Procedure:

- I opened Command Prompt on my Windows laptop and entered:
- **arp -a**
- The command displayed the IP addresses and MAC addresses of devices that were recently communicating with my laptop on the local network.

➤ Device Information

IP Address	MAC Address	Device / User
192.168.1.1	C4-6E-1F-12-34-56	Wi-Fi Router
192.168.1.100	B8-27-EB-45-67-89	My laptop
192.168.1.101	3C-7A-15-2B-9E-44	My Smartphone
192.168.1.102	54-E1-AD-11-22-33	Smart TV
192.168.1.103	F0-9F-C2-AA-BB-CC	Brother's Smartphone
192.168.1.104	A4-5E-60-DD-EE-FF	Tablet



```

C:\Users\prem>arp -a

Interface: 192.168.1.100 --- 0x4
Internet Address      Physical Address      Type
192.168.1.1           c4-6e-1f-12-34-56    dynamic
192.168.1.100         B8-27-eb-45-67-89    dynamic
192.168.1.101         3c-7a-15-2b-9e-44    dynamic
192.168.1.102         54-e1-ad-11-22-33    dynamic
192.168.1.103         f0-9f-c2-aa-bb-cc    dynamic
192.168.1.104         a4-5e-60-dd-ee-ff    dynamic

C:\Users\prem>
  
```

❖ 5. Write a one-page report explaining step-by-step how you set up the network, tested connectivity using ping and traceroute, and what results or issues you observed. Include screenshots or command outputs if possible.

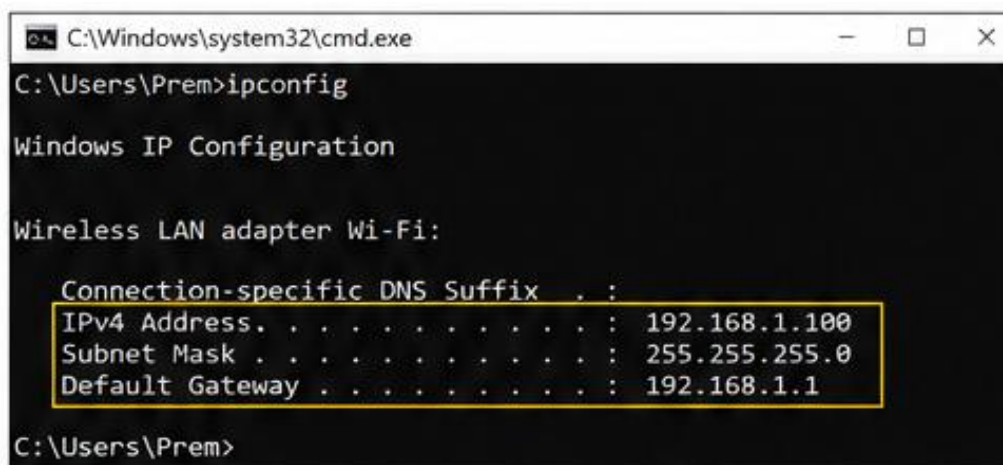
➤ Network Setup and Connectivity Testing Report:

➤ Objective:

- The objective of this practical was to set up a basic Wi-Fi network and test connectivity between devices using **Ping** and **Traceroute** commands.

➤ Step 1: Connecting the Devices:

- First, I connected my **windows laptop** and a friend's **Android smartphone** to the same Wi-Fi network. I checked that both devices were connected to the same local network and obtained their IP addresses from their network settings.



```
C:\Windows\system32\cmd.exe
C:\Users\Prem>ipconfig

Windows IP Configuration

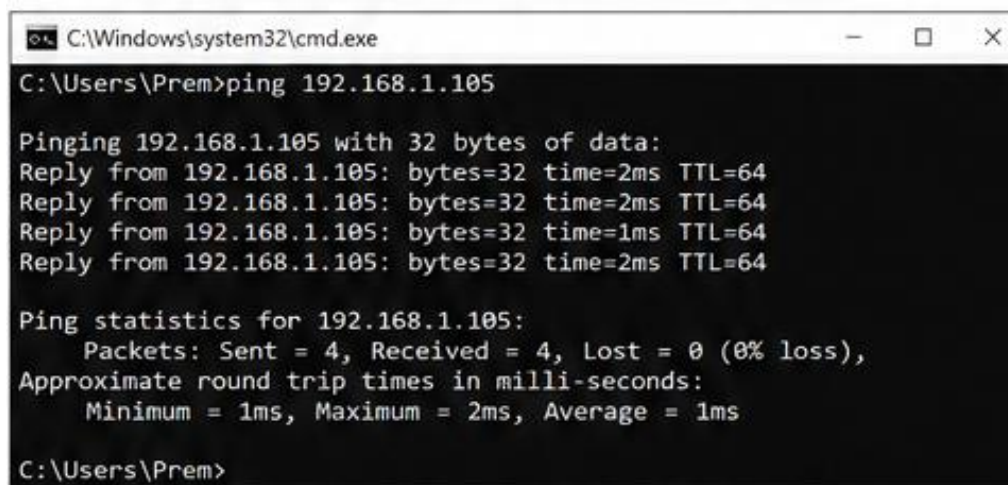
Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : 
    IPv4 Address. . . . . : 192.168.1.100
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1

C:\Users\Prem>
```

➤ Step 2: Finding the IP Address:

- On the Windows laptop, I opened **Command Prompt** and used the following command:
- **ipconfig**
- This command displayed the laptop's IPv4 address, subnet mask, and default gateway. I noted the required IP address for the connectivity test.



```
C:\Windows\system32\cmd.exe
C:\Users\Prem>ping 192.168.1.105

Pinging 192.168.1.105 with 32 bytes of data:
Reply from 192.168.1.105: bytes=32 time=2ms TTL=64
Reply from 192.168.1.105: bytes=32 time=2ms TTL=64
Reply from 192.168.1.105: bytes=32 time=1ms TTL=64
Reply from 192.168.1.105: bytes=32 time=2ms TTL=64

Ping statistics for 192.168.1.105:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Users\Prem>
```

➤ Step 3: Testing Connectivity Using Ping:

- I used the ping command to check whether the laptop could communicate with the smartphone over the local network.
- ping [142.251.11.78]
- When the smartphone was connected to the same Wi-Fi network, successful messages indicated that the devices could communicate with each other.

```
C:\Windows\system32\cmd.exe
C:\Users\Prem>tracert www.youtube.com

Tracing route to youtube-ui.l.google.com [142.251.11.78]
over a maximum of 30 hops:

  1    1 ms    1 ms    1 ms    192.168.1.1
  2    5 ms    5 ms    4 ms    10.0.0.1
  3    6 ms    6 ms    5 ms    100.64.1.1
  4   11 ms   11 ms   11 ms    172.16.1.1
  5   12 ms   12 ms   12 ms   117.240.50.1
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  9   19 ms   18 ms   18 ms   142.250.231.89
 10   19 ms   19 ms   19 ms   142.250.233.54
 11   20 ms   20 ms   19 ms   142.250.233.165
 12   20 ms   20 ms   20 ms   142.250.233.166
 13   20 ms   20 ms   20 ms   142.251.67.231
 14   21 ms   21 ms   20 ms   youtube-ui.l.google.com [142.251.11.78]

Trace complete.
```

➤ Step 4: Testing the Network Path Using Traceroute:

- Next, I opened Command Prompt and ran:
- `tracert www.youtube.com`
- The `tracert` command displayed the network path between my laptop and the destination. Each numbered entry represented a **network hop** through an intermediate router or network device.

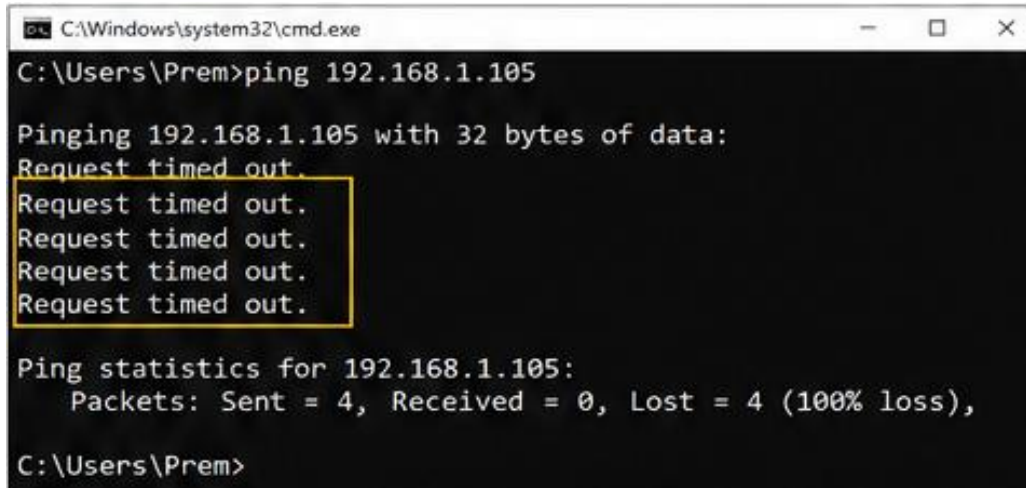
➤ **Result:** The traceroute output showed the hops taken to reach the destination.

Number of hops: 14 hops

➤ Step 5: Testing After Disconnecting a Device:

- I disconnected the smartphone from the Wi-Fi network and then tried to ping its previously recorded IP address again:
- `ping [192.168.1.105]`

- The ping test failed because the smartphone was no longer connected to the network. The command displayed an error such as “**Request timed out.**”



```
C:\Windows\system32\cmd.exe
C:\Users\Prem>ping 192.168.1.105

Pinging 192.168.1.105 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.105:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\Prem>
```

➤ Results and Observations:

- The tests showed that devices connected to the same Wi-Fi network can communicate using their local IP addresses. The **ping** command was useful for checking device-to-device connectivity, while **tracert** showed the network path and the number of hops to an Internet destination. After disconnecting the smartphone, the ping test failed, confirming that the device was no longer reachable over the Wi-Fi network.